

Ahamed Mohamed Wajibu

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EDUCATION

UNIVERSITY OF WATERLOO

Bachelor of Mathematics

Waterloo, ON

May 2028

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, TypeScript, C, C++, SQL

ML & Systems: PyTorch, LangChain, Ollama, RAG

Frameworks/Tools & etc.: FastAPI, ASP.NET Core, React, Node.js, WebGPU, PostgreSQL, SQL Server, MongoDB, Docker, Linux, Azure, AWS, Flask, FastAPI, Node.js, React, Selenium, OpenCV, Nginx, Git, Agile/Scrum, Unit Testing

PROFESSIONAL EXPERIENCE

Mitsubishi

Mississauga, ON

AI Engineer

January 2026 – May 2026

- Built and deployed an internal **RAG platform** over **2,000+ aircraft technical manuals**, combining **LangChain**, **Azure AI Foundry**, **Ollama**, **PostgreSQL**, and **Microsoft Fabric** to support **40+ engineers** at launch.
- Designed an **LLM ETL pipeline** with cross-page context to extract structured fields, summaries, and source metadata into **PostgreSQL**, powering chat search and verification workflows.
- Served **Gemma** and **quantized open-source models** through **on-prem Ollama APIs** behind **Nginx**, keeping document inference inside the corporate network for secure retrieval and analysis.
- Extended the assistant into a **tool-using agent** for document search, page retrieval, **PostgreSQL** keyword search, web search, and inline source citations across technical manuals.
- Developed **ASP.NET Core / React** internal pages for document ingestion, validation, editing, usage reporting, pipeline monitoring, and AI output quality control.

Mitsubishi

Mississauga, ON

Software Developer

September 2025 – December 2025

- Shipped **ASP.NET Core / C# production pages** replacing manual **SQL Server** workflows for QA call-ups, aircraft line status, job updates, and Syspro service operations.
- Developed **SQL** scripts and **ETL** data workflows for production data fixes, validation checks, employee/job records, and manufacturing system updates
- Restored a broken **UiPath-to-Jira production workflow** after an **Atlassian API change** by patching the legacy activity package, then migrating the integration to direct **HTTP** requests for long-term reliability.
- Refactored **Entity Framework** services, **LINQ** queries, and **SQL Server** access patterns, cutting request latency by **~30%** while removing redundant database calls across production modules.

D2DAnalytics

Mississauga, ON

Software Developer

November 2025 – December 2025

- Rebuilt authentication around **OAuth 2.0 / OpenID Connect** with **Microsoft Entra ID**, implementing sliding sessions and seamless reauthentication across frontend and backend.
- Built an **archival and indexing backend** for **X API** data, preserving historical content beyond the platform's native one-week access window for downstream search and analysis.

PROJECTS

Block-GTQ WebGPU | TypeScript, WGSL, WebGPU, PyTorch

- Ported **Block-GTQ's** RoPE-aware KV-cache quantization to **WebGPU/WGSL**, implementing mixed-width encoding, bit-packed storage, and **compressed attention** for browser LLM inference.
- Integrated live compressed K caching into **SmolLM2-135M**, reducing persistent K memory **2.24x at 4K context**.
- Reworked compressed attention to avoid full-K reconstruction, reducing scratch buffers from **2.20 MB to 101 KB** and cutting attention latency by **22x** versus the initial compressed implementation on a 4Kx128 synthetic workload.

BrushAI | Krita Python API, PyQt, Gemini API, Fine-Tuning, Image Inpainting

- Built a **selection-based AI image editor** with **Krita/PyQt**, turning user-selected regions into masks for prompt-driven inpainting while preserving the surrounding artwork.
- Integrated edits into Krita's native layer workflow through its Python **API**, enabling localized object removal and style-preserving transformations with existing undo/redo behavior.